

Relational Distinguishability in a Minimal Non-Neural Carrier

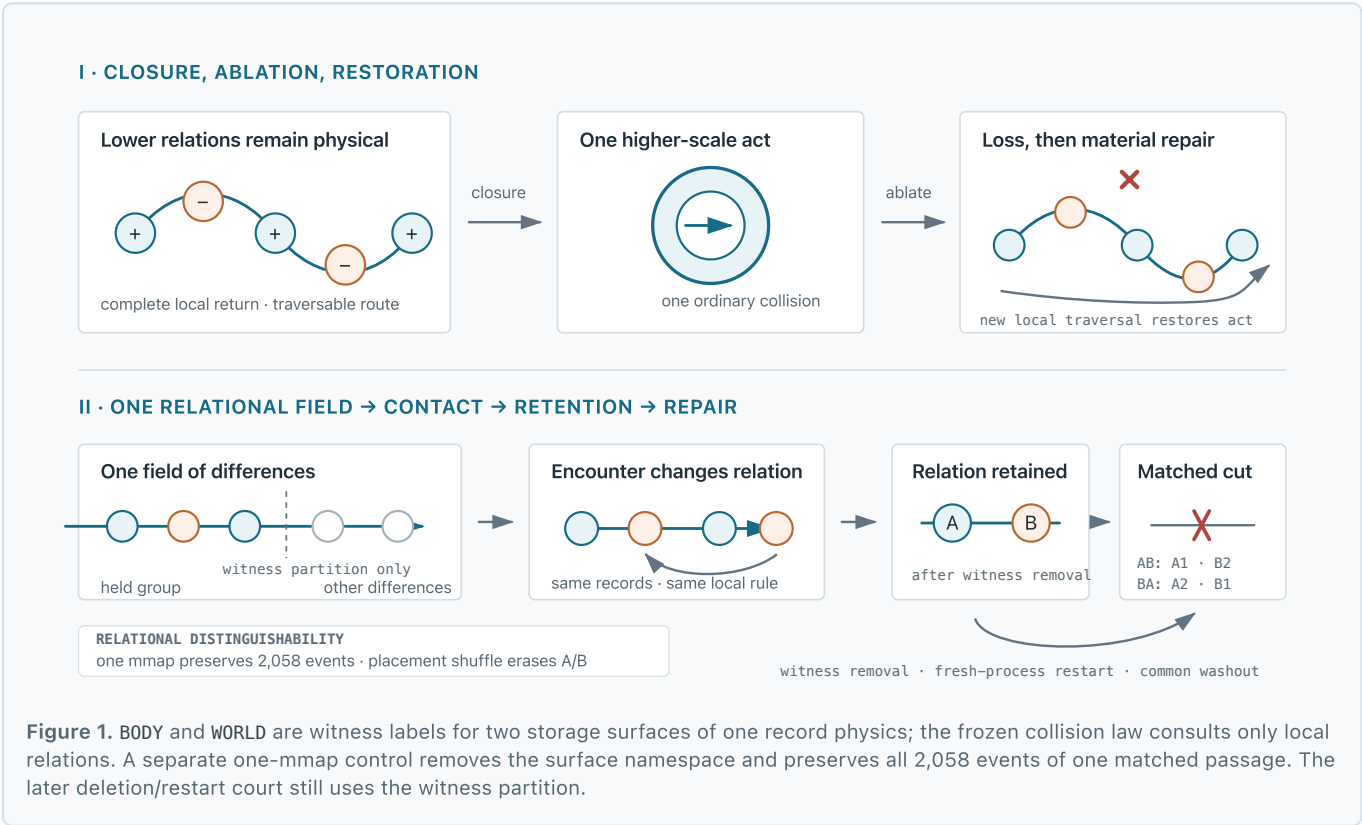
An intervention test through history-dependent repair

Amir Tlinov Independent research July 2026 materials & source two-page PDF

THE BIOLOGICAL QUESTION AND PROPOSED POSTULATE

Work on anatomical homeostasis and basal cognition treats morphogenesis as collective problem-solving in morphospace: tissues can recover large-scale organization after perturbation, while physiological networks retain history not reducible to current anatomy alone [1–4]. **Can later repair depend on history retained in the relational organization of the same substrate that conducts repair?**

Relational Distinguishability (proposed postulate): “No relation is physically meaningful unless its sides are distinguishable.” Here, two histories count as physically different only if their difference changes later repair, survives relation-preserving address relabeling, and disappears when the relevant relational placement is shuffled. The link to relativity is programmatic, not a derivation: the tested invariance is storage-address relabeling, not a spacetime transformation.



OPERATIONAL CRITERION

A relational history must remain distinguishable in future conduct under intervention—not merely classify a trace.

The harness fixes the A/B return opportunity before execution; it does not select a relation online.

1. One local continuation crosses a witness-declared storage partition.
2. A surrounding record must change before return can be earned.
3. The witness removes those records; the retained group restarts.
4. Matched washout and the same new injury follow.
5. Flat/relabel controls must preserve; placement shuffle must erase.

1 rule

flat trace 2,058 / 2,058
collision kernel
0d32047eecb3...

THE CARRIER

The retained group contains 1,024 writable records with explicit local relation handles. The harness calls this storage surface BODY and a temporary five-record surrounding surface WORLD; both use the same record format and collision rule. The frozen kernel has no body/world predicate: surface namespace, lifecycle, injury, and readout belong to the harness. A/B placement is fixed before the run. There is no global sweep, coordinate geometry, target register, reward, planner, or score-dependent update. Frozen audit: 22 tests · world-lineage source SHA-256: d3ed74383729...

History-dependent repair

A and B start from the same injured 513-record retained group. Equal-resource surrounding surfaces differ only in where an earned final return enters two prepared routes. Both recover 1,024 records and the same current upper conductance. After witness removal of the surrounding records, fresh-process restart, identical washout, and the same cut, A restores the upper entry in episode 1 of AB and 2 of BA; B gives the mirror pattern.

Interventions on relational distinguishability

Full address relabeling with relations preserved retains the group's relational organization and future A/B behavior. Shuffling all 257 candidate return-relation placements erases A/B while preserving their multiset, material extent, current upper conductance, and other relation classes. A separate control moves all 1,029 records into one direct mmap with plain, hostilely permuted handles: all 2,058 matched events and all final records remain isomorphic. Separately, upper-relation ablation changes one-collision PASS to HALT; one surviving-cycle traversal rebuilds it.

A1 · B2

upper-entry birth episode in
AB; mirrored in BA

4 / 4

prespecified deterministic
initializations; not
independent replicates

112

paired G1/G2 world-
continuity controls

8 + 8

placement-shuffle + full-
relabel interventions

What the result supports

A bounded constructive result: A/B histories remain relationally distinguishable after removal of the temporary surface, restart, matched washout, and renewed injury. Return-relation placement is interventionally necessary for that distinction, while relation-preserving address relabeling preserves it. The interventions identify a distribution-dependent relation class, not a small memory locus.

What remains unresolved

The witness partition, topology, closure affordance, routes, injury, and readout are engineered. The flat control covers one matched passage, not deletion, restart, washout, controls, or G2 in one mmap. The carrier does not originate, maintain, or defend its own boundary. The prepared organization may be a rudimentary distributed target or only a path-dependent attractor. This is not a biological model, independent replication, or evidence of basal cognition.

BIOLOGICAL QUESTION

Would you regard the topology-encoded closure affordance and history-bearing relational state as a **minimal distributed pattern memory or homeostatic target, only a path-dependent attractor, or as both?** What single perturbation would best distinguish their physical roles?

REFERENCES

1. Levin, M. (2022). *Technological Approach to Mind Everywhere*. doi:10.3389/fnsys.2022.768201
2. Durant, F., et al. (2017). *Long-Term, Stochastic Editing of Regenerative Anatomy via Targeting Endogenous Bioelectric Gradients*. PMID:28538159
3. Manicka, S., & Levin, M. (2019). *Modeling somatic computation with non-neural bioelectric networks*. doi:10.1038/s41598-019-54859-8
4. Levin, M. (2023). *Bioelectric networks: the cognitive glue enabling evolutionary scaling from physiology to mind*. doi:10.1007/s10071-023-01780-3